Travis Dutton

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EDUCATION

University of Maryland, College Park — Expected May 2028
Bachelor of Science, Mechanical Engineering — Cumulative GPA: 3.57

EXPERIENCE

Chassis Engineer — September 2026 – Present
UMD Loop University Rover Challenge (URC) Team — College Park, MD

- Maximize top of chassis surface area relative to internal space by reducing chassis height by 50%
- Perform calculations based on potential failure modes to ensure a robust rover chassis
- Coordinate with electrical, arm, and science subteams to resolve system integration issues
- Produce a SolidWorks CAD assembly for an arm controller, prioritizing design for manufacturability

Undergraduate Researcher - CAD Lead — September 2025 – Present
Bioinspired Advanced Manufacturing Lab — College Park, MD

- Lead two undergraduate researchers in designing components of a soft robotic hand
- Iterate three microchip and fluidic actuator designs with SolidWorks CAD and LCD 3D printing
- Revise manufacturing methods of microfluidic transistors to improve performance based on testing
- Test a novel 3D printing method for microfluidic diodes which use a free-floating disk to control flow

Mechanical Engineering Intern — May 2026 – August 2026
Intralox — Baltimore, MD

- Saved over 100 engineering hours annually by automating design workflows with Autodesk iLogic
- Fabricated a conveyor mount that allows first-attempt alignment for its corresponding transfer
- Designed a bolt placement tool that eliminated sharp sheet metal contact during conveyor assembly
- Utilized automation software Tacton to configure customized conveyor belt designs for customers

PUBLICATIONS

Campbell, P., … **Dutton, T.**, et al. (2026). Fabrication of 3D-Nanoprinted Microfluidic Transistors via Ex Situ Direct Laser Writing. 2026 Hilton Head 22nd Solid-State Sensors, Actuators and Microsystems Workshop.

Campbell, P., … **Dutton, T.**, et al. (2026). 3D Nanoprinting Microfluidic Diodes Via Ex Situ Direct Laser Writing. 2026 IEEE 39th International Conference on Micro Electro Mechanical Systems (MEMS).

PROJECTS

Project Team Member | I-Beam Project — March 2026 – May 2026

- Designed a wooden I-Beam to withstand a point load in a universal testing machine
- Achieved a strength to weight ratio of over 2300:1 by performing calculations with Python
- Compared predicted failure load to actual failure load of 1520 lbf and identified failure mode

Design Team Member | Over Terrain Vehicle (OTV) Project — February 2025 – May 2025

- Designed and constructed an OTV to reach the end of an obstacle course and collect data
- Modeled several vehicle components with CAD and created a full assembly for the OTV
- Analyzed component performance to continue iterative design improvements after initial OTV build

SKILLS & RELEVANT COURSEWORK

Electronic Instrumentation: Multimeter, Oscilloscope, Power Supply, Breadboarding
Data Analysis & Programming: Python, MATLAB, Autodesk iLogic
CAD, PLM & Simulation: SolidWorks, Autodesk Inventor, Fusion 360, SolidWorks FEA, PTC Windchill
Manufacturing: Laser Cutting, 3D Printing (FDM and LCD), Rapid Prototyping
Coursework: Statics, Electronics & Instrumentation, Thermodynamics, Statistics, Fluid Mechanics